

CHAMBERS'S JOURNAL

OF

POPULAR

LITERATURE, SCIENCE, AND ART.

Fourth Series

CONDUCTED BY WILLIAM AND ROBERT CHAMBERS.

No. 724.

SATURDAY, NOVEMBER 10, 1877.

PRICE 1½d.

THE ROYAL GAME OF GOLF.

For ages golf has been pre-eminently the national game of Scotland. As its history emerges from the mists of antiquity we find football and it linked together as representative games, in fulminations against 'unprofitabill sportis,' unduly distracting the attention of the people from more serious affairs. But our game far exceeds this old rival in interest; and if it were not for the popularity of curling in its season, no rival pastime could pretend to vie with golf in Scotland.

The mode of playing golf is so well known in these days that it may suffice to explain that it is a game played over extensive commons, or 'links' as they are termed; that the implements used are peculiarly constructed clubs, so weighted at the crook or 'head' of the shaft, as to give great impetus to the small hard gutta-percha ball to be driven along the grass; and that the object of the players—either as single antagonists or two against two—is to endeavour to vie with each other as to who shall drive the ball towards and into a series of small artificially made holes, in the fewest strokes. From hole to hole the party proceeds, sometimes one winning a hole, sometimes another, and occasionally (by evenly contested play) halving: until the whole round of the green has been traversed; when the party who has gained the greatest number of holes is declared the winner. The links ought to be of considerable extent, and the holes several hundred yards apart, so as to give opportunity for skilful driving and other niceties of the game. To those unfortunates who have only read of the pastime, it may appear hard to believe in the reality of the enthusiasm shewn by its votaries; but whenever they are privileged to come under its influence, even as spectators, they will find it is one of the most fascinating of pursuits. How can a man describe in fitting language the subtle spell that brings him out in all weathers and seasons, and makes him find perfect pleasure in 'grinding round a barren stretch of ground, impelling a gutta-percha ball before him, striving to land it in a succession

of small holes in fewer strokes than his companion and opponent,' as the game might be described by one of that class of men to whom the 'primrose by the river's brink a primrose is, and nothing more.'

The fascinations of the game have enlisted in the ranks of its votaries men of all classes, many of them famous on other fields, who have made their reminiscences of their beloved pursuit mediums for many a bright word-picture in prose and verse. Hitherto no attempt has been made to gather together what has been so said and sung in praise of the pastime; but in Mr Robert Clark's beautiful volume now before us, entitled *Golf—a Royal and Ancient Game*, ample amends have been made for this neglect, by one of the most enthusiastic and best golfers of the day. Here we have presented in a gossiping way so beloved by golfers, wealth of material, both as regards the history and literature of the fascinating game—a labour of love in an artistic guise. What the author is on the links, so seems he to be among his printers and artists and binders—*facile princeps*. The volume before us, though unfortunately too costly to be very generally available, is a marvel of beautiful typography and tasteful binding. Our author has gone for his information to the most various sources—old acts of the Scots parliament, proclamations by kings, burgh records, minutes of the more prominent golf-clubs, books and magazines; and by judicious editing of this medley has shewn the many-sidedness of the game in a way that none but a devotee could.

Mr Clark wastes no space on unprofitable speculations as to the origin of golf. All that is clear in this vexed subject is that though Scotland is the chosen home of the game, she is not its birth-place. It is, however, of little moment whether the game came in with the Scandinavians who settled on the east coast of Scotland, or whether it was brought northward over the Border as a variety of the English 'bandy-ball;' or even if we have to go back to the Campus Martius, and look for the parent of golf in the curved club and feather ball of the Roman *Paganica*. Games of ball seem

CHAMBERS'S JOURNAL

POPULAR

LITERATURE, SCIENCE, AND ART.

Fourth Series

CONDUCTED BY WILLIAM AND ROBERT CHAMBERS.

No. 724.

SATURDAY, NOVEMBER 10, 1877.

PRICE 1½d.

THE ROYAL GAME OF GOLF.

For ages golf has been pre-eminently the national game of Scotland. As its history emerges from the mists of antiquity we find football and it linked together as representative games, in fulminations against 'unprofitable sports,' unduly distracting the attention of the people from more serious affairs. But our game far exceeds this old rival in interest; and if it were not for the popularity of curling in its season, no rival pastime could pretend to vie with golf in Scotland.

The mode of playing golf is so well known in these days that it may suffice to explain that it is a game played over extensive commons, or 'links' as they are termed; that the implements used are peculiarly constructed clubs, so weighted at the crook or 'head' of the shaft, as to give great impetus to the small hard gutta-percha ball to be driven along the grass; and that the object of the players—either as single antagonists or two against two—is to endeavour to vie with each other as to who shall drive the ball towards and into a series of small artificially made holes, in the fewest strokes. From hole to hole the party proceeds, sometimes one winning a hole, sometimes another, and occasionally (by evenly contested play) halving; until the whole round of the green has been traversed; when the party who has gained the greatest number of holes is declared the winner. The links ought to be of considerable extent, and the holes several hundred yards apart, so as to give opportunity for skillful driving and other niceties of the game. To those unfortunates who have only read of the pastime, it may appear hard to believe in the reality of the enthusiasm shewn by its votaries; but whenever they are privileged to come under its influence, even as spectators, they will find it is one of the most fascinating of pursuits. How can a man describe in fitting language the subtle spell that brings him out in all weathers and seasons, and makes him find perfect pleasure in 'grinding round a barren stretch of ground, impelling a gutta-percha ball before him, striving to land it in a succession

of small holes in fewer strokes than his companion and opponent,' as the game might be described by one of that class of men to whom the 'primrose by the river's brink a primrose is, and nothing more.'

The fascinations of the game have enlisted in the ranks of its votaries men of all classes, many of them famous on other fields, who have made their reminiscences of their beloved pursuit mediums for many a bright word-picture in prose and verse. Hitherto no attempt has been made to gather together what has been so said and sung in praise of the pastime; but in Mr Robert Clark's beautiful volume now before us, entitled *Golf—a Royal and Ancient Game*, ample amends have been made for this neglect, by one of the most enthusiastic and best golfers of the day. Here we have presented in a gossipy way so beloved by golfers, wealth of material, both as regards the history and literature of the fascinating game—a labour of love in an artistic guise. What the author is on the links, so seems he to be among his printers and artists and binders—*facile princeps*. The volume before us, though unfortunately too costly to be very generally available, is a marvel of beautiful typography and tasteful binding. Our author has gone for his information to the most various sources—old acts of the Scots parliament, proclamations by kings, burgh records, minutes of the more prominent golf-clubs, books and magazines; and by judicious editing of this medley has shewn the many-sidedness of the game in a way that none but a devotee could.

Mr Clark wastes no space on unprofitable speculations as to the origin of golf. All that is clear in this vexed subject is that though Scotland is the chosen home of the game, she is not its birth-place. It is, however, of little moment whether the game came in with the Scandinavians who settled on the east coast of Scotland, or whether it was brought northward over the Border as a variety of the English 'bandy-ball'; or even if we have to go back to the Campus Martius, and look for the parent of golf in the curved club and feather ball of the Roman *Paganica*. Games of ball seem

The Project Gutenberg eBook of Chambers's Journal of Popular Literature, Science, and Art, No. 724

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Chambers's Journal of Popular Literature, Science, and Art, No. 724

Author: Various

Editor: Robert Chambers
William Chambers

Release date: December 29, 2015 [eBook #50787]
Most recently updated: October 22, 2024

Language: English

Other information and formats: www.gutenberg.org/ebooks/50787

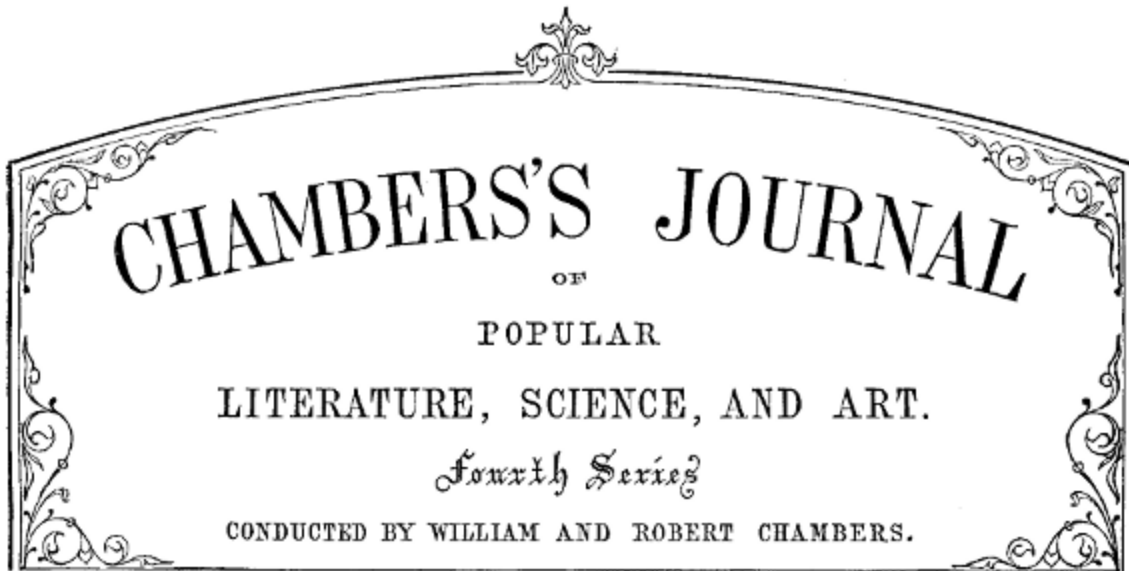
Credits: Produced by Susan Skinner and the Online Distributed
Proofreading Team at <http://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK CHAMBERS'S
JOURNAL OF POPULAR LITERATURE, SCIENCE, AND ART, NO. 724

**CHAMBERS'S JOURNAL
OF
POPULAR
LITERATURE, SCIENCE, AND
ART.**

CONTENTS

THE ROYAL GAME OF GOLF.
FROM DAWN TO SUNSET.
CHARLES DICKENS' MANUSCRIPTS.
THE ADMIRAL'S SECOND WIFE.
THE GUACHO.
THE GERM THEORY AGAIN.
OCEAN-VOYAGES IN SMALL BOATS.



No. 724. SATURDAY, NOVEMBER 10, 1877. PRICE 1½d.

THE ROYAL GAME OF GOLF.

FOR ages golf has been pre-eminently the national game of Scotland. As its history emerges from the mists of antiquity we find football and it linked together as representative games, in fulminations against 'unprofitabill sportis,' unduly distracting the attention of the people from more serious affairs. But our game far exceeds this old rival in interest; and if it were not for the popularity of curling in its season, no rival pastime could pretend to vie with golf in Scotland.

The mode of playing golf is so well known in these days that it may suffice to explain that it is a game played over extensive commons, or 'links' as they are termed; that the implements used are peculiarly constructed clubs, so weighted at the crook or 'head' of the shaft, as to give great impetus to the small hard gutta-percha ball to be driven along the grass; and that the object of the players—either as single antagonists or two against two—is to endeavour to vie with each other as to who shall drive the ball towards and into a series of small artificially made holes, in the fewest strokes. From hole to hole the party proceeds, sometimes one winning a hole, sometimes another, and occasionally (by evenly contested play) halving: until the whole round of the green has been traversed; when the party who has gained the greatest number of holes is declared the winner. The links ought to be of considerable extent, and the holes several hundred yards apart, so as to give opportunity for skilful driving and other niceties of the game. To those unfortunates who have only read of the pastime, it may appear hard to believe in the reality of the enthusiasm shewn by its votaries; but whenever they are privileged to come under its influence, even as spectators, they will find it is one of the most fascinating of pursuits. How can a man describe in fitting language the subtle spell that brings him out in all weathers and seasons, and makes him find perfect pleasure in 'grinding round a barren stretch of ground, impelling a gutta-percha ball before him, striving to land it in a succession of small holes in fewer strokes than his companion and opponent,' as the game might be described by one of that class of men to whom the 'primrose by the river's brink a primrose is, and nothing more.'

The fascinations of the game have enlisted in the ranks of its votaries men of all classes, many of them famous on other fields, who have made their reminiscences of their beloved pursuit mediums for many a bright word-picture in prose and verse. Hitherto no attempt has been made to gather together what has been so said and sung in praise of the pastime; but in Mr Robert Clark's beautiful volume now before us, entitled *Golf—a Royal and Ancient Game*, ample amends have been made for this neglect, by one of the most enthusiastic and best golfers of the day. Here we have presented in a gossipy way so beloved by golfers, wealth of material, both as regards the history and literature of the fascinating game—a labour of love in an artistic guise. What the author is on the links, so seems he to be among his printers and artists and binders—*facile princeps*. The volume before us, though unfortunately too costly to be very generally available, is a marvel of beautiful typography and tasteful binding. Our author has gone for his information to the most various sources—old acts of the Scots parliament, proclamations by kings, burgh records, minutes of the more prominent golf-clubs, books and magazines; and by judicious editing of this medley has shewn the many-sidedness of the game in a way that none but a devotee could.

Mr Clark wastes no space on unprofitable speculations as to the origin of golf. All that is clear in this vexed subject is that though Scotland is the chosen home of the game, she is not its birthplace. It is, however, of little moment whether the game came in with the Scandinavians who settled on the east coast of Scotland, or whether it was brought northward over the Border as a variety of the English 'bandy-ball;' or even if we have to go back to the Campus Martius, and look for the parent of golf in the curved club and feather ball of the Roman *Paganica*. Games of ball seem to have existed in all ages, and it is therefore probable that golf is a development of some older game, or perhaps a 'selection of the fittest' from several previously existing ball-games. It is sufficient for our purpose that early in the fifteenth century it was at least as popular with all classes as it is to-day.

When gunpowder made archery a thing of the past, the conflict between love of country and love of golf ceased, and the game went on prospering under the smiles of royal favour, surviving proclamations of various town-councils directed against sacrilegious golfers whose sin was held to be, not so much that they played on Sunday, as on that part of the day called 'the

tyme of the sermonnes.' This matter was set at rest by the decree of James VI. of Scotland, who in 1618 sent from his new kingdom of England an order that after divine service 'our good people be not discouraged from any harmless recreation,' but prohibiting 'the said recreations to any that are not present in the church, at the service of God, before their going to the said recreations;' or as Charles I., when subsequently ratifying this order, puts it, 'having first done their dutie to God.'

Besides James VI.'s crowning act of founding the Royal Blackheath Club, Mr Clark has recalled two other instances of royal connection with the game in a charming way, as one of the illustrations in his book is from Sir John Gilbert's picture of Charles I. receiving, during a game on Leith Links, the intelligence of Sir Phelim O'Neill's rebellion in Ireland in 1642; while another is a delicately drawn pen-and-ink sketch by Mr James Drummond, R.S.A., of the house in the Canongate of Edinburgh, which John Patersone, shoemaker, built for himself with half the stake in that famous 'foursome'—the Duke of York (James VII.) and Patersone against two English noblemen.

With the Stuarts went out for a time royal countenance of the game, till William IV. became patron of the Royal and Ancient Club of St Andrews, and presented to it for annual competition that coveted golfing trophy, the gold medal.

But though there came kings who knew not golf, the game lost none of its old popularity. Still, as before, pre-eminently the game of the people, we find it associated with many a notable scene and character in the history of Scotland. So fond of the game was the great Montrose, that hardly had the minstrels ceased to serenade him and his day-old bride 'Sweet Mistress Magdalene Carnegie,' when we find him hard at work with clubs and ball. That fifty years later it continued to be the favourite amusement of the aristocracy of the Scottish capital, we can gather from the curious books of expenditure of Sir John Foulis of Ravelstoun, who seems to have spent most of his leisure time 'losing at golfe' on Musselburgh and Leith Links with Hamilton and Rothes and others of the highest quality of the time. We read of Balmerino's brother, Alexander Elphinston, and Captain Porteous, the victim of the famous 'mob,' playing in 1724 'a solemn match at golf' for twenty guineas on Leith Links, where, a few years later, might constantly be seen Lord President Forbes of Culloden, who was such a keen golfer, that when Leith Links were covered with snow he played on the sands; though

even he has to yield in all-absorbing devotion to the game to Alexander M'Kellar, 'the Cock o' the Green,' immortalised in Kay's *Portraits*, who played every day and all day long, and then practised 'putting' at the 'short holes' by candle-light.

It is almost superfluous to say that in our own day the noble and ancient pastime is still the game of the Scots, and latterly of the English, of all classes and in all parts of the world. One little fact that incontestably proves the eminent respectability of the game is that 'the minister' can be a golfer without the least fear of the straitest-laced of presbyteries. It is said that when the canny Scot abroad 'prospects' for a new settlement, while he naturally rivets one eye on the main chance, with the other he reckons up the capabilities of the ground for his favourite game; therefore it is that golf has taken firm root and flourishes in many a distant colony. Across the Border the game is so acclimatised that formidable rivals to our native players are now trained on well-known English greens. That it may go on and prosper is of course the wish of every true lover of the invigorating pastime.

Mr Clark gives us some historical notes of the more prominent of the many golfing clubs that now flourish in different parts of Scotland, and extracts from their minute-books the leading events of their career. Now and then we come across eccentricities, such as the feats of Mr Sceales and Mr Smellie of the Edinburgh Burgess Club in driving balls over the dome of St Giles's Cathedral, one hundred and sixty-one feet high; or the even more wonderful achievement of another member of this club, who drove a ball in forty-four strokes from *inside* their golf-house on Bruntsfield Links over the hill of Arthur Seat. As a rule, however, these clubs pursue the even tenor of their way, the members finding their best happiness in playing the pure and simple game.

While the Honourable Company of Edinburgh Golfers is generally held to be the oldest Scotch Club, so great has been the development of its sister Club at St Andrews, and so great are the attractions of golfing on the famous links of the venerable city, that the 'Royal and Ancient' takes precedence over all, and is indisputably *the* club of the kingdom. What Newmarket is to racing, or Melton to hunting, St Andrews is to golf. In St Andrews, it is not a mere pastime, but a business and a passion. It is the one recreation of the inhabitants from the Principal of the College to the

youngest urchin; it has even invaded the domain of croquet, and has taken captive the ladies, who now take so keen an interest in the game, that on more links than those of St Andrews their green is a charming feature of the place. In short, in St Andrews 'no living thing that does not play golf, or talk golf, or think golf, or at least thoroughly knock under to golf, can live.'

The chief prize of the 'Royal and Ancient'—the gold challenge medal played for every autumn, presented in 1837 by King William IV.—is termed the 'Blue Ribbon of Golf.' To win it is the dream of every member of the Club. Other clubs, such as North Berwick, Musselburgh, Montrose, Perth, Prestwick, Burgess, &c. have each its own time-honoured challenge trophy, that of the Royal Musselburgh being laden with more than a century of medals commemorating each winner. That English clubs too are following fast the fashion set by their older brethren north of the Tweed, is attested by the prizes now competed for at Westward Ho! in Devonshire, Hoylake in Cheshire, and at Wimbledon, &c.; though it is but fair to state that Blackheath claims with good reason to be father of all English golf-clubs, and has for long been celebrated for the keenness of its players and the prizes offered for competition.

So much for the history of the game; let us now glance at its literature. In the interesting collection of prose papers Mr Clark has gathered from various quarters, we can study the peculiar features of the game and the effect it has, for the time, on the tempers of its votaries. As we have seen at St Andrews, the ardent golfer has little time for thought or conversation unconnected with the game. For the time being the be-all and end-all of his life lies within the pot-hook-shaped course he has to traverse; and not a little of his happiness or his misery for the day depends on the nature of the match he succeeds in getting. Though the game is as a rule an exceedingly social one, and admits of quiet chat and occasional good-natured banter, the *true* golfer at work is essentially a man of silence; chattering during the crises of the game is as abhorrent to him as conversation during whist; one thing only is as obnoxious as the human voice to him then—that is, any movement of the human body near him. 'Stand still while I'm putting,' and 'Don't speak on the stroke,' are two postulates he would fain enforce. This over-sensitiveness to external influences may explain the seeming ungallantry of the 'Colonel' in H. J. M.'s amusing account of *The Golfer at Home*, which appeared in the *Cornhill Magazine* a few years ago. After a

charming little picture of the 'Colonel' resenting, though he does not openly object to Browne being accompanied over the course by 'his women,' as he ungallantly terms Mrs Browne and her sister, he says to his partner: 'The Links is not the place for women; they talk incessantly, they never stand still, and if they do, the wind won't allow their dresses to stand still.' However, as they settle down to their game, the 'Colonel's' good temper returns under the healthy influence of an invigorating 'round,' and gives H. J. M. an opportunity of pointing out how all ill-humours of body and mind give way before the equable and bracing exercise of a round or two of the Links of St Rule. That the reader may see the amount of walking exercise taken in a round of St Andrews Links, it may be interesting to note that the exact distance, as the crow flies, is three miles eleven hundred and fifty-four yards; so that the golfer who takes his daily three rounds walks *at least* eleven miles. It is no wonder, then, that in addition to its own attractions, golf is esteemed as a capital preparation for the moors or the stubbles, hardening as it does the muscles both of arms and legs. What hunting does for the cavalry soldier as a training for more important bursts in the battle-field, the like does golf for the infantry soldier in bracing him to encounter forced marching with ease. The Links have formed the training-ground of many a brilliant officer.

Space will not allow us to dwell on the genial gossip about St Andrews and St Andrews players—amateur and professional—that we find in Mr Clark's book, further than to mention three names. First, that of the great champion of the professionals, Allan Robertson, who was 'never beaten in a match;' of the brilliant but short-lived career of poor 'young Tom Morris,' the champion player of his day—son of a worthy sire who still survives; of Mr Sutherland, an old gentleman who made golf the chief business of his life, whose interest in his fellow-men, not as men but as golfers, is well shewn in this anecdote. His antagonist was about to strike off for the finishing hole at St Andrews, when a boy appeared on the bridge over the burn. Old Sutherland shouted out: 'Stop, stop! Don't play upon *him*; he's a fine young golfer!'

It is in verse, however, that the votary of golf finds the field congenial to his subject.

In 1842 appeared a clever collection of poems, entitled *Golfiana*, by George Fullerton Carnegie of Pittarrow, which delighted the golfers of that day by

the humorous way in which it hit off the playing characteristics of the men he introduced into it. He begins by throwing down the gauntlet to those students of Scottish history who sigh over the musty memories and deplore the decayed glories of the city of their patron saint:

St Andrews! they say that thy glories are gone,
That thy streets are deserted, thy castles o'er-thrown:
If thy glories *be* gone, they are only, methinks,
As it were by enchantment transferred to thy Links.
Though thy streets be not now, as of yore, full of prelates,
Of abbots and monks, and of hot-headed zealots,
Let none judge us rashly, or blame us as scoffers,
When we say that instead there are Links full of golfers,
With more of good heart and good feeling among them
Than the abbots, the monks, and the zealots who sung them!

We have many capital songs in honour of the game; amongst others a parody of Lord Houghton's well-known song, *Strangers yet*, from which it will be seen that something more is necessary to make a good golfer than a set of clubs and an anxious 'cady' to carry them:

DUFFERS YET.—BY TWO 'LONG SPOONS.'

After years of play together,
After fair and stormy weather,
After rounds of every green
From Westward Ho! to Aberdeen;
Why did e'er we buy a set
If we must be duffers yet!
Duffers yet! Duffers yet!

After singles, foursomes—all,
Fractured club and cloven ball;
After grief in sand and whin,
Foozled drives and 'putts' not in—
Ev'n our cadies scarce regret
When we part as duffers yet,
Duffers yet! Duffers yet!

After days of frugal fare,
Still we spend our force in air;
After nips to give us nerve,
Not the less our drivers swerve;
Friends may back and foes may bet,
And ourselves be duffers yet,
Duffers yet! Duffers yet!

Must it ever then be thus?
Failure most mysterious!
Shall we never fairly stand
Eye on ball as club in hand?
Are the bounds eternal set
To retain us duffers yet?
Duffers yet! Duffers yet!

In conclusion, we may remark that though golf, to the uninitiated, may appear to be a game requiring considerable strength of muscle for its achievement, it is not so; for the easier it is played, the better are the results. To apply much force to the stroke is to imperil the chance of driving a far ball; whereas by a moderate swing of the club, the ball is not only driven far and sure, but goes from no effort apparent to the striker.

A notion also prevails that golf is a game suited for young and middle-aged folks only. This is a delusion, for no outdoor pastime is more fitted for elderly people. To attain *great* excellence in the game, the player must commence early in life; but to become enamoured of its joys requires but a beginning, and that beginning may be made by men who have long passed the meridian of life. We could point to many elderly gentlemen whose lives are being lengthened by the vigour-inspiring game, and who, when their daily round or rounds are finished, can fight their battles o'er again in the cheery club-house, with all the zest of youth. When games such as cricket have been found too much, or perhaps the exertion of tramping the moors too severe, the sexagenarian may safely take to the easy but invigorating pursuit of golf, and 'bless the chiel who invented it.' If he misgives his ability to cope with the exertion, or fancied exertion, of pacing a few miles of green turf and wielding a club, our advice to him is to place himself in the hands of a professional golf-player—plenty of whom are to be found

wherever there are links—and try; and in a wonderfully short time our veteran may find himself interested, perhaps absorbed, in a game the delights of which he has lived all these years without having been able till now to realise!

FROM DAWN TO SUNSET.

PART III.

CHAPTER THE SECOND.

DEBORAH waited and watched—a gloom unutterable weighed on her spirits—and no Mistress Fleming came. At last old Jordan Dinnage arrived at the castle alone, looking scared and sorrow-stricken.

'The master is very ill,' said Mistress Marjory, as she waited on Jordan. 'These be bad days, Master Dinnage. I doubt if he lives till morning. Doctor says he won't; but doctors know naught. In general, if doctors say "He'll be dead by mornin'," it means he'll live to a good old age; I've seed it often. But mark my words, Jordan Dinnage: there's not much life in our dear Master; *he's goin'*. This comes o' leavin' Enderby. I felt it; I knew'd 'twould be so. *This comes o' Master Sinclair's leavin's*. O Jordan Dinnage, it's wrong, it's grievous wrong, this leavin' Enderby, for this grand blowed-out old place, an' these flaunting livery-men an' maids. Master Sinclair's curse is on us!'

'Nay, nay, Mistress Marjory; these be women's superstitions. Mistress Deborah did rightly. A goose she would ha' been to fling all this grandery and gold guineas in the ditch, for fear o' bad luck, 'sooth! It's no more that, than thou'rt a wise woman! The Master'll pull through; an' if he don't, better die a prince than a beggar.'

Marjory shook her head. 'Give me honest beggary. An' where's Mistress Dinnage? Be sure Lady Deb 'ud be glad o' her company now. Why didst not bring her along, Jordan? It speaks not much for her love.'

Jordan reddened. 'Not a word agen Meg, Mistress Marjory! She'll be comin' soon. I must see Mistress Deborah.'

'Well, come now. An' heaven send Master Kingston soon.'

Deborah met the dear old man with outstretched hands. 'Jordan, I am so glad to see ye! Where is Margaret?'

Jordan shuffled from one foot to the other, and twisted his hat round in his hands. 'Well, Lady Deb—Mistress Deborah—I've not brought Meg along.'

'I see ye have not!' cried Deborah impetuously. 'But where is she?'

The old gray eyes, growing dim with age, looked straight and honestly on their young Mistress, yet humbly too, as he answered in a low voice: 'Where she ought to be, Mistress Deborah—off to her young husband, Master Charlie Fleming.'

'Jordan, Jordan! Is this true? Her husband? Ye bewilder me. Are they wedded then? Is she gone to Ireland?'

'Sure enow! O Mistress Deborah, I come to ask forgiveness! It isn't for the like o' Jordan Dinnage to have his daughter Mistress Fleming; but dear heaven knows I know'd naught, an' never sought it out, nor had high notions. Mistress Deborah, I ask forgiveness, an' I hope the master'll forgive me.'

Deborah took the old trembling hand. 'The master is in no state to blame or to forgive. But, Jordan, thou may'st give me joy o' this. It gladdens mine heart in my sore troubles like a sunbeam on a dark, dark cloud. Forgive thee? Ay, I am proud to be Margaret Fleming's sister; an' well believe my father would bid her welcome too—faithful honest Jordan. Now come, Jordan, come, and see how he lies. He knows me not, and he calls ever upon Charlie. Hast sent my letter to Ireland? Hast the address?'

'Ay, ay; it's gone.'

'Then I will write again to-night. Heaven send he may come in time. Sometimes, Jordan, he lieth in a stupor; again he calls for Charlie or for me.'

Reverently pulling his white forelock, with his old habit of respect, to his fiery but beloved master, Jordan stood at the foot of the bed, and saw the shadow of death on the face of Vincent Fleming.

'My boy,' murmured the dying man, with his eyes upon Jordan—'my boy Charlie!'

Old Jordan gazed helplessly and sorrowfully from him to the doctor who stood by, and Marjory, who entered. 'What's to be done?' he muttered. 'It kills him!'

'Patience, patience!' whispered the solemn doctor; 'he may see his son yet. There is great hope for him, Mistress Fleming; keep good heart.'

'Not hope of his recovery, Master Allan,' said Deborah, with stern and still despair. 'I know death when I see it. You have held out hope before; yet make him live till my brother comes. Ye hear me, Master Allan?'

'Ay, Mistress Fleming; I will use my poor skill to the utmost. Bear up. I will return to-night, Mistress Fleming;' and with a courtly bow, he left her.

But for Deborah, she kneeled beside her father, and with old days and old memories her heart was like to break. Jordan was weeping bitterly; she heard the old man's sobs; but on her own heart a still Hand was laid, enforcing strength and calmness. For two things she prayed: that Charlie might come in time; and that her father might be himself before he died, to hear that Charlie had ever been true to him. And so through the long night she watched; and old Marjory oft slept and nodded, as age and dulled senses will; and though Sir Vincent at times called plaintively for his Deb, his 'Rose of Enderby,' his more frequent plaint was for his boy.

CHAPTER THE THIRD.

In those days there were wild doings in Ireland. 'Liberty and Reform' were the watch-words which did then, and ever will, electrify the fiery, rebellious, ardent spirits that flocked under one banner to struggle and to die. Irish and French met and fought together against the iron hand of England; thousands perished; the fated isle ran blood.

It is the eve of a battle. Gray dawn is slowly breaking over forest and mountain, where strange and wonderful echoes are wont to be heard amongst the rocks and caves; but in the gray of this dread dawn, on the eve of battle and blood, all seems silent as the grave, saving the thunderous roar of the waterfall in its descent into the lake, that seems to make the silence the more intense.

But hark! through the mist of morning a bugle suddenly sounds loud and clear; and when it ceases—far away, a spirit-bugle answers. A soldier, driven to frenzy, they say, by an insulting taunt from a superior officer, had struck him down in the heat of the instant. Short shrift in those days; the man has been tried, condemned, and is about to be led out to execution. So, loud and clear the bugle calls: 'Come forth to thy death,' as plain as a human voice could speak; and he whom it summons cannot mistake that voice, and comes forth guarded, but with steady step, and head erect and soldierly; while in front of him bristles a long line of musketry, and behind yawns an open grave. The condemned soldier is Charles Fleming. Have his ungovernable passions and his strong uncurbed will brought him to this? Ay; and the stubborn pride which has ever been his bane, leads him now to die without that word of extenuation or appeal which even yet might save him.

Yet who may tell how that proud heart swells well-nigh to breaking beneath the broad breast, as he thinks on the old white-haired father and his son's death of shame! He sees too the shadows on the woods of Enderby. He hears the voice of a little sister, calling 'Charlie, Charlie!' at play. And the trees are waving their long arms round the old, old home; and his little playmate Margaret—his *young wife Margaret*—stands beneath and smiles. And then his bold eyes ask for death, merciful death, which shall put him out of his anguish. Yet hold! Even as the muskets are raised, but ere the triggers are pressed, there is a wild shriek of 'Rescue! rescue! Pardon! pardon for Charles Fleming!'

And there, headlong down the way—while all reel back before him—rides one spurring for life or death, his horse in a lather of foam, his head bare, and his long hair flying in the wind. In one hand he clenches a packet, and waves it above his head—the Royal pardon! He reaches them; he stays the deadly fire with his wild outstretched arms raised to heaven, with white face and blazing eyes, and lips which fail to speak. But *one* could have undertaken and accomplished that famous ride; but *one* could have saved him in this strait. In male disguise, that *one* proves to be Margaret Dinning! 'Tis my wife!' cried Charles Fleming in piercing accents; 'tis my wife Margaret!' And with that, the king's messenger sways in the saddle, and is supported to the ground by the commanding officer....

And thus it came to pass that Deborah, watching at her father's bedside, heard rumours of that battle by which the name of Charles Fleming became famous. It was early morning. The great wild clouds of dawn were parted, and rolled asunder. The glorious sun rose on the watcher's weary eyes, and steeped the land in splendour. Deborah threw up the windows wide, and returned to the dying man. O heaven, tender mercy, cannot the light of summer sunrise rob that dear face of aught of its wintry wanness?

'Father, sweet father!' she said in thrilling tones of grief, 'art thou not better? See the glorious sun, father!'

'Nay, Deb,' he answered plaintively; 'I see no sun; mine eyes are dark. How little thou dost look to me! Thou'rt grown so small! My child, my darling, I am very ill.'

Then Deborah raised his head upon her shoulder; she knew that he was himself again, himself but to die; her brave heart sank, yet she answered calmly: 'Yes, thou hast been very ill. Dost thou remember all that happened?'

'Ay, ay. My boy, my boy!' And he sobbed.

'Hush, father; that was wrong; that was false! That was a wicked forgery. Charlie never wronged thee by thought or deed. Charlie hath ever been loyal to thee and thine. Art thou content now, dear?'

A brilliant smile stole over the fading face of Vincent Fleming. 'Ay,' said he, 'content to *die*!' He lay musing, his eyes closed. 'Deb,' said he at last, 'whisper me. My boy is true to me—is't not so?'

'Yes, father; true as steel: he loves thee dearly. And for *thee*,' she went on, with heaving breast, 'he hath done brave things! Charlie is a soldier, and men are all saying he hath won great honour and renown.'

'Ah, Deb; thank God, thank God for this! And thou, Deb, sweet Deb, how is't with thee?'

'I am rich, dear. I am betrothed to King Fleming, whom I love most dearly; and I have wealth enough for all. It is well with thy two children, thou seest.'

And ere the night fell, two messengers came gently to his side. One, radiant with 'white raiment' and drooped wings; the other, footsore, travel-stained, and war-worn. And one was the Angel of Death, who stood and looked

upon them pitifully; the other was his prodigal son, who kneeled and folded his arms around his father, and bowed his head and wept.

'Now,' said Sir Vincent, 'I die in peace. How have I yearned for thee! God bless thee! I bless thee, my boy! Deb, this is death!'

And so, raised in Charlie's strong arms and with his hands in Deborah's, without a struggle, the spirit passed away.

CHAPTER THE LAST.

Two figures stand together in one of the deep oriel windows of the old hall at Enderby. The blood-red splendour of a setting sun fills the marsh, the low land, and the hanging woods; and streaming like a beacon in at the windows, floods those two with radiant light. They are Charles Fleming and his bride. The storms have swept by, and left her thrice his own, with the old walls and the sacred hearth of Enderby. Thus may God send on us the lightning of His chastisement, and yet guide and guard us through all—through the morning of wild and sunny childhood; through the noon of gay and love-bright youth, environed as it is by perils; through the sudden-falling night of dread, despair, and death. He does not leave us 'comfortless.' As for Deborah Fleming, passionately as she loved the beauteous world, she never again lost sight of the valley up which had passed the souls of those she loved, and the golden gates across the shining flood. And in later days, when children's children clustered eagerly round the stately old Lady of Lincoln, she, with the faithfulness of old age, would return lovingly, lingeringly to the days of her youth, when 'Charlie and she were young.'

O happy time—blessed childhood—how can I end better than with thee? Over the shadows of evening rises the day-star of childhood's memories.

It knows no night—
There is *no* night in a glad and green old age.

THE END.

CHARLES DICKENS' MANUSCRIPTS.

A GLIMPSE of the manuscripts of the late Charles Dickens, which now form part of the 'Forster Collection' in the South Kensington Museum, conjures up a vision of numerous characters in his popular novels. On looking attentively at the manuscripts, we are at once struck by the number of alterations and interlineations with which the pages abound; and our first sentiment is one of surprise that the books which appear so wonderfully natural and fluent when we read them, should evidently have been the result of much anxious thought, care, and elaboration.

The collection comprises the original manuscripts of the following works: *Oliver Twist*, published in 1838-39; *Master Humphrey's Clock*, comprising the *Old Curiosity Shop* and *Barnaby Rudge*, published in 1840-41; *Barnaby Rudge*, a separate volume, 1840-41; *American Notes*, 1842; *Martin Chuzzlewit*, 1843-44; *The Chimes*, Christmas 1844; *Dombey and Son*, 1846-48; *David Copperfield*, 1849-50; *Bleak House*, which has in the original manuscript a secondary title, *The East Wind*, 1852-53; *Hard Times*, 1854; *Little Dorrit*, 1855-57; *A Tale of Two Cities*, 1859; and *The Mystery of Edwin Drood* (his last but unfinished work), 1870. There are also proof volumes from the printers, consisting of *Dombey and Son*, *David Copperfield*, *Bleak House*, and *Little Dorrit*, the pages of which bear marginal and other corrections and alterations, in ink, by the author.

Of course, as the collection is placed under a glass case, the public can only see one or two pages of each work; but even with this meagre guide, the acute observer is able in some degree to trace the working of the writer's mind, and to follow to some extent the development of his ideas. As we have already remarked, the first thing which strikes us is the comparatively large number of alterations and interlineations which occur in the manuscript. It is evident that Charles Dickens wrote with the greatest care, and scrupulously revised his writing, in order to render each sentence as perfect as might be. Taking the works in their chronological order, we may notice that in *Oliver Twist*, which is open at 'Chapter the Twelfth'—'In which Oliver is taken better care of than he ever was before, with some

particulars concerning a certain picture'—there are few alterations in the manuscript; the writing also being larger and firmer than in the majority of the later works. Charles Dickens made his alterations so carefully that it is difficult to trace the words which he had originally written; but the one or two which occur on this page give us some little insight into the careful manner in which the author worked up his sentences into a well-rounded and euphonious form. The passage at which this manuscript is opened runs as follows: 'The coach rattled away down Mount Pleasant and up Exmouth Street—over nearly the same ground as that which Oliver had traversed when he first entered London in;' and here occurs the first alteration, 'the D ——' is erased, and 'company with the Dodger' is written in its place; the author evidently considering the latter a more euphonious form of expression than 'in the Dodger's company,' as it was doubtless his original intention to make the passage. The alteration to which we have referred may appear, as indeed it is, of exceedingly small significance; but we have mentioned it simply as an instance of the extremely careful way in which Dickens studied the details and minutiae of composition.

The next manuscript in point of date is *Master Humphrey's Clock*, which is open at 'No. IV.,' headed 'Master Humphrey from his clock-side in the chimney corner,' and commences as follows: 'Night is generally my time for walking. In the summer I often leave home early in the morning and roam about fields and lanes all day, or even escape for days or weeks together, but, saving in the country' [this originally stood 'but, at other seasons of the year;'] but Dickens doubtless saw that the expression as it now stands would be more consistent with the context], 'I seldom go out until after dark, though, Heaven be thanked, I love its light and feel the cheerfulness it sheds upon the earth as much as any creature living.' This page of manuscript has only a moderate share of alterations.

Then we come to the volume of *Barnaby Rudge*, which is opened at 'Chapter One,' and also contains only a moderate number of alterations, one being in the height of the *Maypole* sign, and another in the distance of Epping Forest from Cornhill; both of which are noticeable as further illustrations of the conscientious love of accuracy which characterised the author's mind. Next in order follows the *American Notes*, which has very few corrections, and is opened at the page headed 'Chapter the First. Introductory and necessary to be read;' in which the author challenges the

right of any person 'to pass judgment on this book or to arrive at any reasonable conclusion in reference to it without first being at the trouble of becoming acquainted with its design and purpose.' Surely a caution fair and reasonable enough on the part of the writer of a book which he could not but feel would probably give offence, where such an end was farthest from his wish.

The Life and Adventures of Martin Chuzzlewit comes next, open at 'Chapter I. Introductory. Concerning the Pedigree of the Chuzzlewit Family;' and giving us a brief but telling satire on the pride of birth by assuring us that this family 'undoubtedly descended in a direct line from Adam and Eve, and was in the very earliest times closely connected with the agricultural interest.' This page is notably full of alterations, and seems a fair indication that with Charles Dickens, as with many others, the first step was the most difficult of all. The caligraphy in this as in all the other manuscripts is legible but rather small, the letters being distinctly formed, and the use of abbreviations studiously avoided.

We next turn to *The Chimes*, one of those delightful stories with which Dickens introduced to us those Christmas annuals, which now form so important a section of our periodical literature. This again is open at the commencement, where the author lays down the dogma that there are not many people who would care to sleep in a church: 'I don't mean at sermon-time in warm weather (when the thing has actually been done once or twice), but in the night, and alone.' This sentence originally finished with 'in the night;' but we can readily imagine the development of the idea in the brain of the writer; and the words 'and alone' suggesting themselves as lending an additional ground of fear for the situation. The manuscript of this page bears a moderate number of alterations.

In *Dombey and Son* we find a large number of alterations on the first page, the very title itself having been altered more than once. The sketch of the newly-born Paul, who was placed in front of the fire, 'as if his constitution were analogous to that of a muffin, and it was essential to toast him brown while he was very new,' is very good indeed; but it is evident that the passage was rather the result of careful elaboration than of spontaneous humour. And the same remark will apply to the opening chapter of *David Copperfield*, in which, although the passage descriptive of the birth of the

hero is very neat and natural as it now stands, the same careful revision and alteration are again apparent.

Bleak House too is notably full of alterations on the first page, especially in the passage which tells us that in the muddy condition of the London streets 'it would not be wonderful to meet a Mesalosaurus forty feet long or so waddling like an elephantine lizard up Holborn Hill.'

In *Hard Times*, where we are introduced to the gentleman who wants nothing but 'Facts,' and in the opening chapter of *Little Dorrit*, in which we have a description of Marseilles as it 'lay broiling in the sun one day,' we find a large number of alterations; but in these, as in most of the other instances, the primary words have been erased so carefully, that it is next to impossible to form an idea of how the passages originally stood. The *Tale of Two Cities*, on the contrary, contains remarkably few corrections; and the opening passage descriptive of 'The Period' is telling, and apparently written spontaneously. *The Mystery of Edwin Drood* has been opened with good judgment at the last page. The manuscript is very small, but fairly legible, and having but a moderate number of alterations. In a literary sense, it is not perhaps so interesting as some of the others; but it possesses a sad and melancholy claim upon our attention and sympathy, inasmuch as it is the last page of manuscript ever written by this gifted hand.

In the proof volumes with corrections in the handwriting of the author there is nothing which calls for especial note save an unimportant deletion in *Bleak House*, and a more interesting alteration in *David Copperfield*. In the former there is a passage marked 'out,' in which Sir Leicester Dedlock speaks to Mrs Rouncewell of her grandson in the following passage: 'If (he said) the boy could not settle down at Chesney Wold, in itself the most astonishing circumstance in the world, could he not serve his country in the ranks of her defenders, as his brother had done? Must he rush to her destruction at his early age and with his parricidal hand strike at her?'

In *David Copperfield* we find by a passage in which Mr Dick is referring to his Memorial that his original hallucination took the form of a 'bull in a china shop;' a rather trite idea, and it was not until after the proof had actually been submitted to him by the printers that Charles Dickens introduced the whimsical and happier notion of 'King Charles's Head.'

Before bringing our brief paper to a conclusion, we would venture to suggest to the gentleman or gentlemen to whom is intrusted the arrangement of these manuscripts, that the present positions of the manuscripts and printed volumes should be transposed, so that the manuscripts should occupy the lower half of the case, as in their present position it is rather difficult to decipher the caligraphy; and to any one below the ordinary height it must involve an amount of physical contortion as uncomfortable as it is inelegant. The manuscripts being of course of greater interest than the printed proofs, should certainly occupy the more prominent space, especially as the latter could be read without any difficulty if placed in the rear rank.

We have no doubt that many of those who read this short article will have seen the Dickens manuscripts for themselves; many more doubtless will see them; but there will still be a large number who will not have the opportunity; and while we think that our remarks will be endorsed by the first and second classes, we hope that they will prove interesting to the third less fortunate class, and will enable them to enjoy, at least in imagination, a somewhat closer intimacy than they have known before with that great and gifted man, whose books have effected so many beneficial changes both in society at large and in many an individual heart and life, uprooting and casting to the winds much that was base, worthless, and contemptible, and implanting in their stead the seeds of those gentler sympathies and nobler aspirations which find their fruition in a well-spent life.

THE ADMIRAL'S SECOND WIFE.

CHAPTER XII.—OTHER EVENTS OF THAT EVENING.

LADY DILLWORTH'S reverie is doomed to be a short one. She feels a soft caressing touch on her arm, and looks up to see Miss Delmere close by her chair. Her long light hair is streaming over her shoulders, and an embroidered Indian dressing-gown covers her antique dress.

'Liddy, you quite frightened me! Why do you come creeping in like a mouse? You ought to be in bed.'

'I have something to tell you, Katie; something you will be so glad to hear, and something that makes me so happy. I cannot sleep till I tell you all about it.'

Miss Delmere flings herself on a low stool at Katie's feet, and looks up through her mass of sunny hair with flushed cheeks, glowing eyes, and lips that *will* form themselves into smiles. She cannot hide her joy.

'Walter Reeves has asked me to be his wife. Are you surprised, Katie?'

'Not exactly; I thought there must be some outcome from all that flirting. Do you know, Liddy, if he had not made you an offer, and if you had not accepted him, I should have been very angry, and should have given you a lecture.'

Liddy looks up at her friend with surprise, the words are so cold, the tone of voice so hard and unsympathising.

'Are you not glad about it, Katie?'

'Of course I am; and I hope you will both be happy.'

'I owe it all to *you*, darling Katie! Had it not been for this dear delightful charade party, I should never have found out that Walter really cared for me. How sudden it has all been! And what good news I shall have to carry home to-morrow! Little did I think when I came to stay with you, that my

wedding was so near!' The words came out in joyous gasps between hugs and kisses, for Miss Delmere is demonstrative, and shews it.

Then Liddy flits away, radiant in her delight, never dreaming of the anguish in Katie's heart that constrains her again to bury her face in her hands, and utter short, eager, impassioned prayers for the poor sailors whom she believes are at that very hour in dire and mortal conflict with the winds and waves.

But we must take a glimpse at Sir Herbert's proceedings. He never even glances at the order after his wife's fingers have altered it to her will; he merely folds it up, puts it in the envelope, and despatches it to its destination. Though he decides the *Leo* shall proceed on the dangerous enterprise, no thought of malice towards Captain Reeves actuates him. It never enters his thought that it is a good way of getting rid of him for a while, and thus stopping the constant visits to Government House. The idea is altogether too paltry and despicable—it is beneath a man of Sir Herbert's tone of mind. He fixes on that particular ship simply because she is best fitted for the duty. Weighing anchor in such a storm near the Short Reefs on an iron-bound coast, and rendering assistance to a vessel in danger, is an undertaking that requires a good ship, a steady crew, and an able captain.

All these qualifications the *Leo* possesses to perfection. She is a well-built handsome craft; her hardy tars are smart and well disciplined; and there is no braver officer in the British navy than Walter Reeves. True, when on shore he seems rather too fond of amusement, and has been called 'conceited,' 'trifling,' 'frivolous,' 'dandified,' and what not, by men who are jealous of him; but let his foot once touch the quarter-deck, and even his enemies can never charge him with these questionable qualities. There all his frippery and nonsense vanish away like dew in the sunshine; and he becomes the true sailor, with courage to plan and carry out deeds of daring; he becomes the gallant officer fired with vigour and ambition. Never would he shirk a duty or hesitate to undertake any lawful enterprise even though it led to danger or death. Sir Herbert knows all this, and therefore he is right in selecting the *Leo*.

Hardly has he sent away the order when he is called off to Hillview; and when his duties there are over, he determines to pay a farewell visit to Lady Ribson. He thinks of Katie all the way he is going to Belton Park. But when

is he *not* thinking of her? His love has not lessened, though he has begun to see her faults. He is sorry she is not with him, and that she has never paid the needful respect to his god-mother. He has often and often urged her to call, but his persuasions have failed. Whenever he has made the suggestion, Katie has been so overwhelmed with engagements that she has hardly given him a hearing, and of late he has dropped the subject. He goes towards Belton Park in rather a gloomy mood after all. Lady Ribson quite expects Katie on this last evening, and while she welcomes the Admiral, she looks over his shoulder inquiringly.

'Ah! I knew you would come to say "good-bye," Herbert. But where is the "gudewife," the bonnie Katie?'

'Miss Delmere is staying with her, and she has many engagements; besides, you could hardly expect her out in this storm.'

'Ah no, certainly not. There are many reasons for Lady Dillworth's staying at home, and but few inducements for her to come out to see an old woman like me.'

'Katie has often said how anxious she is to know you.'

'True, true, Herbert; so you must bring her to Scotland with you in the bright summer-time—that is, if I'm spared to see it; but life is uncertain, my friend, life is uncertain.'

Lady Ribson, who is the brightest, kindest, dearest old woman in the world, smiles on her god-son, and does not let him see how much she is hurt by Katie's neglect of her; but in her heart she is sorry for him, more sorry than she would like him to know. Bessie his first wife was in her opinion perfection; and Katie she suspects is very much the reverse.

To her old eyes, the Admiral is still young, and she thinks there is hardly a woman in the world good enough to mate with him. 'I can see Herbert is not happy; and Laura Best was right when she foretold the risk her father ran in marrying a mere frivolous girl,' she decides in her own mind; but none of her suspicions float to the surface, so gay, so kindly, so warm is her manner. The Admiral sets out early on his homeward journey; his thoughts still turn to Katie, but they have grown softer, more tender. The gloom has passed from his spirit; the interview with Lady Ribson has calmed his ruffled thoughts; his reserve and pride have altogether melted down, and he longs

to press his darling wife to his heart and forgive all her follies. He feels, even with all her failings, he loves her more completely, more passionately than he has ever loved the dead Bessie.

When he reaches Government House, it is brilliantly lighted up. The guests are assembled, and fragments of song and melody are floating out on the rough night-wind. Sir Herbert makes his way at once to the scene of festivity, and pauses at the door, astonished at the unwonted appearance of the rooms. As he has not been initiated into the arrangements, nor witnessed the preparations, the merest stranger present is not more ignorant than he is of all that has been going on. So he looks on the scene with curiosity. The music-room has been turned into a raised stage, with painted wings and festoons of scarlet curtains. A crescent-shaped row of gas jets serves as foot-lights, and throws a soft clear brilliance on the performers. Wreaths of flowers, clusters of trailing evergreens, pots of rich exotics, groups of banners, add to the display. Nothing that taste, art, fancy, or money can accomplish is wanting. The Admiral looks at all this; then at the rows of spectators; then at his wife, who comes forward on the stage at that moment leaning on Liddy's arm. Presently their voices ring out through the rooms; then a solo falls to Katie's share, and her husband listens spell-bound to her singing. Her voice is tuned to the deepest pathos, and her face is sad as her song.

Never has he seen Katie look like that before. The curiously cut costume suits her wonderfully well; the dress of azure silk falls in rich bright folds; her bodice glitters with gold and gems; and her hair turned back in its own luxuriant wealth of tresses, has no ornament but a diamond cluster. The mellowed rays from gas jets, hidden by the curtains, fall full on her head, and she shines out as though surrounded by a strange unearthly glory.

She seems altered, spiritualised, refined, incorporealised in her marvellously weird-like beauty, and her husband cannot remove his rapt gaze from her. But presently a single turn of his head changes his glance of admiration into one of surprise and anger. In the shade of a gigantic azalea he spies Captain Walter Reeves, standing in an attitude of calm listening enjoyment. Instantly the Admiral's eye flashes with indignation. How dares Walter Reeves to be here, in his wife's drawing-room, when he ought to be miles away out on the stormy seas?

In an instant the offender is called out of the room, and Sir Herbert demands to know why he has disobeyed orders by staying on shore.

'I have had no orders to weigh anchor, Sir Herbert.'

'Perhaps the order is still lying on your cabin table; it was issued at ten o'clock this morning.'

'No despatch has reached the *Leo*, for I've been on board all day, Sir Herbert, and came direct to Government House.'

'Very strange, very! There must be some terrible mistake in the matter. Is Mr Grey here to-night?'

'No, Sir Herbert.'

'I must see him at once. The subject admits of no delay.'

'Shall I go to North Street, and fetch him here?'

The Admiral pauses for a moment, and takes a survey of Walter from head to foot. He notes the velvet suit, the delicate lace ruffles, the Montero cap, the large plume of feathers, the dark cloak set so jauntily on his shoulder, the thin shining shoes, and the huge glittering buckles; and a *soupçon* of contempt glances from his eyes, a slight sneer trembles on his lip. 'I think I am more fitted to brave the storm than you are to-night, so I'll go to Mr Grey myself.' Then without another word, he walks down the stairs, and passes out into the wind and rain. The house in North Street is closed for the night, and Mrs Grey and Helen are sleeping the sleep of the quiet-minded. Only the master of the house is still up, and he is finishing a cigar in his library. He starts up in alarm when he hears the authoritative knock at the door, and visions of fire and thieves start up before him. His alarm is in no whit lessened when he sees his august son-in-law on the steps.

'Sir Herbert! Who would have thought of seeing you so late! Is anything wrong? Is Katie ill?'

'No; your daughter is quite well. I left her just now dressed up like some medieval heroine, and lamenting her woes in song.'

'True; I recollect this is the night of Katie's charade party.'

They have both gone into the library now; Mr Grey has flung the stump of his cigar aside, and the Admiral speedily explains the cause of his late visit.

'I acted as your note directed, Sir Herbert, and at once sent off the *Leoni* to assist the *Daring*.'

'The *Leoni*! Were you mad, Grey?'

'I confess your order amazed me. I did all I could to consult with you about it, but you were gone to Hillview. Here is the order; you will see the *Leoni*'s name written plainly.'

The Admiral takes the paper in his hand, holds it near his gaze, scrutinises it afar off, glances at it through his eye-glass; but the fact is indisputable—there is the word *Leoni*, apparently in his own writing.

'This is a vile forgery, Grey! I never wrote that, never dreamt of giving such a mad order. Heaven alone knows what results, what complications may arise from it! I shudder to think of the *Daring* still aground on the Short Reefs, or perhaps altogether broken up long ere this.'

'The *Leoni* couldn't help her much, I fear.'

'Help her! She'll never reach her. I should not be surprised if she were a wreck herself by this time; a hideous, top-heavy, unmanageable craft like that couldn't take care even of herself in such a storm.'

'What had better be done now, Sir Herbert?'

'Despatch the *Leo* at once; though I fear her services will come too late.'

Practical discussions follow, that keep the Admiral and his secretary employed for some time longer. When Sir Herbert returns home, it is no vain excuse that makes him retire to his room in very weariness of spirit, very fatigue of body. He finds Walter Reeves is already gone away; but some of the guests are still lingering in the rooms, trying to prolong their amusements to the last minute.

CHAPTER XIII.—CONFESSION.

The storm has spent itself before the next morning. Katie can see that, as she listlessly looks out of the bay-window of the breakfast-room. One would hardly suppose the treacherous gale had been holding such wild revels the night before. The tossing waves that had leaped with frothy crests over the serrated rocks of the Short Reefs, are placid enough now—dancing

perhaps over those who went down a few hours before into the cruel depths. Lady Dillworth has a headache; she listens calmly to Liddy, who blushing and blooming, pours forth her rose-coloured confidences, and swallows her coffee between whiles. Hunter is helping the groom to carry her boxes down-stairs; and Miss Delmere, with only a few minutes to spare, is selfish in the exuberance of her joy, and cannot see the dark circles round Katie's sleepless eyes nor note the deep sadness of her looks. At length she goes away, and the Admiral enters the room.

'You are just in time, Herbert; Hunter has brought up some fresh coffee.'

'None for me, thank you. I knew you would be engaged with Miss Delmere; and as I had papers to examine, I had my breakfast brought to the library.'

'Liddy is gone away now.'

'Yes; I met her in the hall, and saw her into the carriage. I've brought you the newspaper, Katie; you will see the wreck of the unfortunate ship I told you of yesterday.'

'The *Daring*! Is she wrecked?' Katie takes the paper into her trembling hands, but cannot read a word for the throbbing of her brows and the dizziness of her eyes.

Her husband goes on: 'Yes; she went to pieces in the gale, and every soul on board would have gone down with her had not a merchant-ship passed by the merest chance. Twenty-three men are lost. At least they went away in the *Daring's* large cutter; but no boat could have lived out the storm.'

'How dreadful!' Katie starts at the sound of her own voice, it is so deep and hoarse.

'Dreadful indeed! What makes the matter worse is, that in all human probability every man might have been saved and the ship also, had not an atrociously wrong act been perpetrated.'

Katie hears a rustle of paper; she knows by instinct what is coming, but she dares not lift her head.

The Admiral goes on in an agitated tone: 'Some one has tampered with my papers, has even dared to meddle with my orders. I directed the *Leo* to be sent out at once to the scene of the wreck; but from malignity or some other motive, the name *Leoni* was substituted.'

'Wouldn't that ship do as well, Herbert?'

'Certainly not. She would never reach the Short Reefs in such a gale. I fully suspect she's foundered at sea or gone on the rocks herself. I'll find out who did it! If I thought Reeves, or any one else at his instigation, had been guilty, I'd, I'd'——

There is no saying how the sentence might have ended. Katie has risen from her seat, and stands before her husband trembling.

'I did it, Herbert! *I* altered your order!'

'You, Katie!—you, my wife!'

'Yes; but I never thought my silly act would lead to such misfortune.'

'What was your motive, Kate? Surely you could not have wished to injure *me*? To set me up as a mark of inefficiency and ridicule?'

'O no; a thousand times *no*. But Captain Reeves was helping me to get up our charade, and I altered the ship's name that he might not have to go away.' Here Lady Dillworth's voice fails her. She cannot utter another word, so choked and gasping is her breath; the bare blank sentence remains as it was: 'I altered the ship's name that he might not have to go away.'

The Admiral does not reply. There is a stillness in the room as though some one had died there. A burst of passion, an angry storm of words would be a relief; and Katie glances up in alarm to see her husband looking down sadly at her. He is pale as death; his lips are set and firm; a dim haze has clouded his eyes, as though unshed tears are springing there; but there is no sign of resentment in his face—only pity, a tender, touching, tremulous pity, an infinite yearning for something gone, a regret, sorrowful and deep! Yet all so mixed with intense love, that Katie knows for the first time in her life what passionate boundless strength there is in his affection for her. A sudden understanding of how dear she is to him dawns upon her; she feels he would give his very life for her.

Katie would have flown to his arms, and told him his love is fully returned, that at last she feels his worth and goodness; she would have fallen at his feet and there have craved for pardon; but he puts her gently yet firmly away.

'My poor, poor Katie! Have I then spoiled your young life? I might have suspected this; but I was blind and selfish. Forgive me, my poor child, forgive me! I would give worlds to restore you your freedom again!'

Ere Katie has fully grasped the meaning of his words, he has gone out of the room; she hears him walk rapidly down the stairs and out of the house. A sense of numbness creeps over her; she sits for a while like one stunned. How long she remains crouching on the sofa she never knows; a whole lifetime of anguish seems crushed into that space. All the brightness of youth appears to die out at her husband's departure; his retreating footstep sounds like a knell of departed hope.

After a time, Lady Dillworth rouses herself; even sorrow cannot endure for ever. She recollects it is near the hour for luncheon, and then Herbert will come home. She dresses herself in the robes she had on when he made her the offer of marriage. *Why* she has done this, she does not confess even to herself; but perhaps she imagines old associations may soften present misunderstandings. She goes down to the dining-room and waits. The table is laid for luncheon, and the bright fire glitters on the silver and glasses and flowers. All is so pleasant and cheerful and homelike! And even then a thrill of satisfaction comes over her that now Liddy Delmere is gone she will be able to devote all her time to her husband—have him all to herself. But the luncheon hour passes, and then the door opens and Hunter enters with a letter on a salver. The address is written in a rapid unsteady hand, as though the fingers trembled. She sees it is Sir Herbert's writing, and tears open the envelope with a sense of impending trouble, that blanches her cheeks and chills her heart. The words run thus:

'No one shall ever know you did the mischief, my poor Katie; the blame shall rest on me alone, and I will bear it willingly for your sake. But my professional career is over; men will never again trust my judgment or deem me fit to command. I was proud of my standing in the service and of an untarnished reputation; but you have spoiled it all, merely to enjoy a short interval more of Walter Reeves's society. Why did you not tell me he was so dear to you? You should have said before we married *I* could never make you happy. Yet I will not blame you, my poor wife. My own selfish blindness has caused all this misery. Before this letter reaches you, I shall be on my way to London to resign my appointment.'

This was all! But the contents fell like a blow on her heart. Katie sits alone in that quiet room while the iron pierces her soul. The untasted luncheon stands on the table till the fire goes out and the shades of night gather round. Then Hunter knocks at the door in alarm, to know if 'my Lady' will have the things removed. Katie rouses herself to tell him that while his master is away she will henceforth have her meals laid in her boudoir, and that she will receive no visitors in Sir Herbert's absence.

Hunter sees her pallid face and tear-stained eyes, and draws his own conclusions, and thinks things 'never went on like that in the *first* Lady Dillworth's time, anyhow.'

THE GUACHO.

'WILL you ride over with me to the neighbouring village?' asked my friend Senhor Pedro da Silva. 'There is a *festa* there to-day. And as you are a stranger in the country, you will see some feats of horsemanship quite as clever as can be shewn in the circus rings of old England.'

'With the greatest pleasure,' I replied. 'I have often heard of the wonderful horsemen called Guachos, and desire much to see if the accounts are really true.'

'I think you will not be disappointed. He and his horse are one; sometimes he acts as its tyrant, but more frequently they are friends. From infancy they have scoured over the immense Pampas of South America, frequently amidst violent storms of thunder, wind, and rain. His address and grace on horseback yield neither to your best fox-hunters nor to the American Indian. But here is Antonio with our steeds; let us mount.'

An hour's ride over the dull arid plains of Buenos Ayres, covered with the grass now so much cultivated in our gardens, and admired for its light leathery tufts waving in the wind, brought us to San Joachim, where the people were already collecting in their holiday attire, and exchanging friendly greetings on all sides. The gay striking dresses of the Guachos mingled in every group. The *poncho* or mantle of cloth, woven in bright coloured stripes, has a hole in the centre through which the head is passed, and falls down to the hips in graceful folds. The nether garment is a combination of bedgown and trousers, bordered by a fringe or even rich lace on these festa days, which varies from two to six inches deep according to the wealth of the wearer. Then to-day the great jack-boots of untanned leather are exchanged for the smartest patent leather, with bright scarlet tops, and enormous spurs at the heels. A wide-brimmed Spanish hat is worn, a purple or yellow handkerchief twisted round it; whilst the belt encircling the waist sparkles with the dollars sewn upon it—often the whole fortune of the owner. His weapons are attached to this girdle, consisting of a formidable knife, a lasso, and a bolas, which may not be so familiar to the English reader as the lasso. There are two balls fastened together by short

leathern straps, to which another thong is attached, by which it is thrown; this is whirled violently round the head before propulsion, and entangles itself in the legs of the horse or cow to be captured.

But whilst we are gaily chatting to Senhor Pedro's many friends the games are beginning, and we hasten off to the ground. There we find two lines of mounted Guachos, from ten to twenty on each side, just so far apart as to allow a rider to pass between the ranks; all are on the alert and holding the lasso ready for use. One whom they call Massimo, an evident favourite with the crowd, comes tearing along at a gallop and dashes in between the lines. The first horseman in the ranks throws his lasso at Massimo's horse as he flashes past, but misses, amidst the derisive shouts of those around; then the second, quick as lightning casts his; and so on down the ranks. Presently, however, the horse is lassoed and brought to the ground; and the skilful rider alights uninjured on his feet, smoking his cigarette as coolly as when he started from the post. The dexterity and watchfulness of the men, who can throw the lasso so as to entangle the feet of a horse while going at full speed, are simply wonderful. Another and another followed with varying fortunes; sometimes the first struck down the horse and rider, rarely was it that one escaped altogether. The popularity of the famous chief Rosas was said to be founded on his proficiency in this adroit but cruel art, and no man can be their chief who is not the cleverest among them: renown on horseback is the one great virtue that exalts a man in their eyes; cruelty to their favourite animal does not seem to enter into their thoughts!

But at length they weary of this sport, and move off a little way to vary it with another. Now we seem to have moved back a few hundred years, and find a pastime of the middle ages still lingering among these descendants of the Spaniards, who doubtless introduced it into the New World. In those days it was called the game of the quintain. A pole was firmly planted in the ground, with a cross-bar, to which was hung the figure of a misbelieving Saracen, well armed and holding a large sword. The horseman tilted at full gallop against this puppet; and as it moved lightly on a pivot, unless it were well struck in the breast, it revolved, and the sword smote the assailant on the back amidst the laughter of the crowd. Here in the wild Pampas the trial of skill is greater. A kind of gibbet is erected, to which is hung a finger-ring by a string. The Guacho, instead of the spear of knightly days, holds a weapon more characteristic of his work in the *saladero*, where the cattle are

killed and salted—namely, a skewer. One after another the Guachos gallop at full speed and try to push the skewer into the ring and carry it off. Antonio, Luis, and Melito succeeded admirably; but many a novice failed in the difficult task. Still it was a pretty sight, and enjoyed apparently by both horses and men.

Then came the inevitable horse-races, which are of almost daily occurrence, when associates challenge one another, and they strike off in a moment in a straight line until they disappear in the horizon. In this case, however, a wide straight avenue near the village was chosen for a short, rapid, and often renewed race; a pastime for the idle, and the occasion of ruinous bets. The riders were dressed with the greatest elegance; their horses well chosen from the corral, and covered with silver ornaments. The bridle is of the leather of a foal, finely plaited and mounted with silver; stirrup, bit, and spurs of the same metal. A glittering silver belt, sometimes of a flowery pattern, and of colossal proportions, hangs round the breast, and a silver strap across the forehead. The saddle is a wonderful piece of mechanism, forming the Guacho's 'bed by night and chest of drawers by day;' it is very heavy, and consists of ten parts; skins, carpets, and cow-hides intermingled with other necessities. Off they go at last from the post, spurring and urging their steeds like modern centaurs, handling them in a manner well worthy of admiration, and with the most perfect elegance. When the winner came in, many a by-stander had lost all his possessions, so mad a race of gamblers are they. As a last resource, they pledge their horse, and expose themselves, if they lose, to the lowest of humiliations—that of going away on foot!

We turned at last towards home, leaving the roystering spirits to finish off their day at the *pulperia*. This it is which takes the place of the club, the café, the newsroom, and the home. A cottage, neither more simple nor more luxurious than any other to be found in the Pampas, covered with thatch; the walls of dried mud, or more frequently of rushes sparged with mud; the flooring being of trodden earth; into which the rain penetrates, the sun never enters, and where a hot damp air is the prevailing atmosphere. Before the door stands a row of strong posts, to which the horses of the guests are tied; the new-comer jumps off, and there leaves his steed, saddled and bridled, for many weary hours in the hot sun or pouring rain; whilst he, to use a native expression, 'satisfies his vices' in the *pulperia*. The door is open to all comers, and great outward politeness reigns within; there is a continual

exchange of gallantries, to which the Spanish language easily lends itself; but reason soon loses its sway, and the strangest bets are offered and taken. Sometimes it is between two friends as to who shall first lose blood; when the whole company sally out, knives are drawn between the duellers, and a combat, often much more ridiculous than valiant, ensues!

The following morning, Senhor Pedro proposed that we should ride out and see the Guacho at work and in his home. 'You seem to have been interested in him yesterday,' he said, 'and he belongs to a type that is unique. Notwithstanding the hatred of the original inhabitants towards their invaders, the two races were mixed, and these unions produced the Guacho. Look at his tall figure, bony square face embrowned by the sun, and stiff black hair—there you see the Indian; whilst the Spaniard is in his proud haughty manner, in his vanity, and also in his great sobriety. He drinks water and eats his dried meat without bread, not from contempt for better food, but from a horror of work. To earn his daily food is not so much his aim as to get money to bet with. He will go into the *saladero*, where, knife in hand, he will kill, skin, and cut up the cattle for salting, and find enjoyment rather than labour in it. He easily gains in a few hours a wage that suffices; and as soon as it is paid, he jumps on his horse and rides off to the *pulperia* to gamble it away.'

Thus conversing, we reached a hut which could scarcely be surpassed in its misery. Placed alone in the middle of the plain, without any garden or cultivated ground, not a tree to cast a welcome shadow, or a hand to repair the dilapidated walls, it seemed formed to repulse rather than attract the owner. At our approach, the mother came out, surrounded by her children, her complexion approaching the mulatto, for the air of the Pampas quickly destroys the fineness of the skin. It is only in the capital, Buenos Ayres, that handsome Creole types are to be seen, where fine features of an Indian class surpass European beauty, even when the tint is olive. The wife, like the husband, hates work: her only occupation is to boil some water, pour it over maté or tea of Paraguay, and drink it through a metal tube. Her children, at the age of three or four, can sit on horseback and gallop over the plain with no other bridle than a cord passed through the horse's mouth. At six they watch the sheep, and at ten are ready to break in the most spirited colts. Only everything they do must be on horseback: they will neither use their arms nor legs.

'Good-morning, Senorita,' said my friend. 'Where shall we find your husband?'

'He is gone, Senhor, to break in some horses for Senhor Melisos; it is not far from here.'

'So much the better. We will ride on and see him at work.'

We reached the place; and the Guacho came out to meet us.

'Will you shew my friend your feat at the gate?' said Senhor Pedro.

'With the greatest pleasure,' answered the flattered Guacho. He jumped on to the top transverse bar which forms the gate of the corral, and calling to another man to open the lower ones and drive out a troop of horses at full gallop, he, with the most astonishing skill, singled one out with his eye, dropped down on to it, and rode off without saddle or bridle at the top of its speed. Soon returning, he proceeded to break a horse that had been previously caught in the plains. The Guacho threw two lassos, one over the neck, the other on the hind-legs. Several men hold the colt tightly whilst he saddles and passes a cord through the mouth of the animal; and when the first paroxysms of fear have passed, the tamer jumps on, and pressing his powerful knees into its sides, the lasso is withdrawn. The horse and rider then start on a furious course, from which they both return exhausted, in the midst of the *vivats* which resound from every side. All that is now required is for the breaker to ride ten or fifteen leagues, when he gives up the horse to the owner and receives his fee. They are never taught to trot, but have an easy movement; and a man has been known to ride two hundred miles a day without fatigue, and living only on dried meat and maté.

THE GERM THEORY AGAIN.

WE have on several occasions alluded to the Germ theory, by which is meant the theory that invisible germs, capable of producing animalculæ and of spreading disease are constantly floating in the atmosphere—and that the more impure the air the greater are the number of these germs. We revert to the subject, because it is debated in all quarters, and it is as well that our readers should know something of what is causing so much controversy. Some surgeons distinguished as operators are great believers in the Germ theory; so much so, that before beginning, for instance, to cut off a leg, they cause a certain germ-killing liquid to shower like spray near the part operated on; by which, as is alleged, the wound is kept free of anything noxious. Whether there be germs or not, the use of disinfectants in the air is said to be beneficial. Notably the celebrated carbolic-acid plan of Professor Lister has met with marked success, and is practised by the greatest surgeons of our time. But though the air certainly contains something which favours decomposition, it is by no means yet proved that that something *is* made up of germs.

Professor Tyndall has been the principal advocate of the Germ theory, and has written some papers strongly in its favour. Professor Bastian takes an opposite opinion. He thinks that living organisms may originate in disease by spontaneous generation. His notions are that if germs are continually floating about in the air, they would drop everywhere and anywhere alike. This argument applies more forcibly to the fact which Dr Bastian discovered—namely, that he was able to get life in flasks containing inorganic solutions, but that he always failed if such solutions were not made up of salts containing oxygen, hydrogen, carbon, and nitrogen; that is to say, of the elements of life. If the organisms are really the result of a molecular arrangement of the 'mother-liquid,' we should expect to find them only in those fluids which already contain the elements necessary for their composition. Three speculations are involved in these experiments: on the one hand, that low forms of life do occasionally arise by spontaneous generation; on the other hand, either that the heat which is usually

considered destructive of life and germinating power is in reality nothing of the kind; or that Dr Bastian's experiments were incorrectly performed.

Since the publication of Dr Bastian's observations, a very lively controversy has been carried on in scientific quarters between the supporters of the germ theory and of the theory of spontaneous generation. Dr Bastian's work was conducted with great care and in the presence of some distinguished authorities. Dr Sanderson, on the other hand, found that upon increasing the heat which is applied to the flasks, no organisms were produced; but until we have reason to doubt the generally received opinion as to the amount of heat necessary to destroy life, this result may be equally well explained according to either of the two theories.

Dr Bastian insists that the organic solutions in his own flasks are not found by him to undergo putrefaction where every precaution is taken for withholding the entrance of air. Thus a simple piece of cotton-wool, which acts as a kind of sieve, will when placed in the mouth of a flask prevent decomposition. Professor Tyndall has invented the most ingenious contrivances for illustrating his views. In one case he employed a chamber the walls of which were covered by a sticky substance. The particles of dust in the air were allowed to collect and adhere to the sides, and the air in the vessel, as shewn by its non-reflection of a beam of light, was rendered comparatively dustless. Flasks were now introduced, and they remained for a long period free from organisms. On repeating some of these experiments this year, however, Professor Tyndall found that many of the infusions which had previously been preserved from putrefaction with ease, were now found, when placed under the same conditions, to swarm with life. Still he refused to believe in 'spontaneous generation,' and preferred to consider that the production of life in his flasks was due to some fault in his experiments, and that the air of the Royal Institution was not so pure this year as it was last. Instead, therefore, of introducing his fluids by means of an open pipette, as he had previously done, he now made use of a 'separating funnel,' and by this means the fluids found their way into the flasks without exposure to the air. The result of these precautions was that no organisms appeared. The objection, however, that we have to find is, that no guarantee can be given that will enable us to ascertain whether the air is really free from particles of organic matter or not. Last year the air was

considered to be pure because moteless; but this year, though moteless, it was found to be impure.

Professor Tyndall and his friends are so exceedingly confident in the power of the germs of the atmosphere, that they attribute to their influence every known case of putrefaction; and they do so because they believe that they have proved that whenever the air can be excluded from a putrescible fluid, putrefaction will not take place. But Dr Bastian has succeeded in producing life out of organic infusions from which the air has been excluded, and which have been previously raised by him to temperatures hitherto considered by scientific authority as fatal to life. Thus the question resolves itself into this: What is the exact point of heat which kills the germs of bacteria? At present we do not know, and we have therefore no right to make any supposition upon this point in favour of either of the two theories.

Since Dr Bastian's experiments were first made public, the holders of the Germ theory have gradually raised what we may call the thermal death-point of bacteria, in order to explain away the results of his experiments by the light of their own theory. If Dr Bastian's fluids did develop life, they say, the germs must have entered into them by some means or other; and if he superheated these fluids, the fact of the germs surviving the process shews that they must be possessed of greater enduring power than we have given them credit for.

Curiously enough, Professor Tyndall declares that frequent applications of a low degree of heat, and applied at intervals, have a far greater 'sterilising effect' than a single application of a high temperature. For a given fluid may contain germs of all ages. If such a fluid be boiled for a considerable time, all the germs of recent formation will be killed; but those of a greater age will merely be softened, but still capable of reanimation. If, however, the fluid be heated for a short time only, the recent germs will be destroyed, while an older crop will be liberated. A second application of heat destroys this second crop, and brings a third into play. Further heat will awaken successive crops, until at length a point is reached when the toughest germ must yield. This is certainly a most ingenious explanation of the difficulty.

A very interesting contribution to this subject has lately been made by Dr Bastian and others; and we will now briefly describe the main results of their researches. It has long been known that slightly alkaline organic fluids

are more difficult to sterilise than those which are slightly acid. Pasteur the French chemist says that animal water in its normally acid state becomes sterile at one hundred degrees centigrade; but that if the infusion is first rendered alkaline by the addition of potash, the application of a little more heat is necessary, in order to insure sterility. If we bear in mind the two theories, we shall see that these observations of Pasteur may be explained according to either of them. We may believe that the germs in the infusion are fortified against the destructive action of heat by liquor potassæ; or on the other hand, we may hold that the spontaneous generation of organisms is favoured by the presence of an alkali. Acting upon these data, Dr Bastian heated a similar fluid in its acid condition to the temperature of one hundred degrees; so that, according to Pasteur, it was now barren. He then added a quantity of potash sufficient to neutralise the acid, the addition of the alkali thus being made *after* instead of before the boiling; and he then allowed the fluid so treated to stand at a temperature of about one hundred and fifteen degrees Fahrenheit. In a short time swarms of bacteria appeared.

Dr Roberts, however, considers that this result was obtained because sufficient precaution had not been taken by Dr Bastian to prevent the entry of germs, which might have been introduced by the potash. Accordingly, he filled a small flask with an ounce of the acid infusion, and then sealed up his potash in a capillary tube. The potash was then heated in oil to two hundred and eighty degrees Fahrenheit, and kept for fifteen minutes. The tube of potash was now introduced into the flask containing the infusion, and the flask was boiled for five minutes, and sealed. The flask was now kept for some time in order to test its sterility. When this was ascertained, the flask was shaken, so that the little tube of potash inside was broken, and the potash was thus allowed to mingle with and neutralise the infusion. The flask was now maintained at a low temperature of one hundred and fifteen degrees Fahrenheit, and it remained perfectly clear. And so Dr Roberts concludes that liquor potassæ has no power to excite the generation of organisms in a sterilised infusion. Professor Tyndall repeated these experiments with additional precautions, and obtained similar results.

The *general* conclusion which is drawn from various experiments by the advocates of the Germ theory is, that liquor potassæ has no inherent power to stimulate the production of bacteria, and that any apparent power of this kind which it may seem to possess is due to the presence of germs within it.

These germs they consider are not destroyed until the potash has been raised to the temperature of one hundred degrees centigrade if solid, and to one hundred and ten degrees centigrade if liquid. Dr Bastian, who repeated his former experiments with every possible precaution, found no difference in his results. Moreover, he discovered that liquor potassæ, when added in proper quantities, is just as efficacious in stimulating the development of life after it has been heated to one hundred and ten degrees centigrade, as when it has been heated to only one hundred degrees. Pasteur will consequently have to raise the temperature which he considers sufficient to destroy the germs contained in a solution of strong liquor potassæ to a point still higher than one hundred and ten degrees.

But there is still another proof that liquor potassæ if previously heated to one hundred degrees does not induce fermentation in virtue of its germs, because if only one or two drops be added, the infusion will remain as barren as ever; while a few more drops will immediately start the process of fermentation. Now if the potash really induced fermentation because it brought germs along with it, two drops would be quite as efficacious as any other amount. Finally, Dr Bastian has shewn us that an excess of alkali prevents fermentation, and to this fact he attributes the failure of Pasteur to develop life when he employed solid potash. He had added too much of the alkali.

It is impossible to draw any *definite* conclusion from these as from the other experiments, until we know the precise temperature which is fatal to germinal life. Dr Bastian indeed thinks that he has been able to shew that bacteria and their germs cannot exist at higher temperatures than one hundred and forty degrees Fahrenheit; but his evidence here is not quite conclusive. He does not deny the existence of germs nor their probable influence in producing life; he merely says that his experiments furnish evidence to shew that in some cases organisms may spring into existence without the aid of a parent. The strong points of his case are, that as fast as his adversaries can suggest precautions to insure the destruction of germs, he has been able to shew life under the altered conditions; and that whenever the supposed death-point of bacteria has been raised on account of his experiments, he has succeeded in obtaining life after having submitted his flasks to the required temperatures.

How this most interesting controversy will end, we cannot foretell; but we hope that the further researches of our scientific men upon the subject will ultimately lead to the discovery of the truth. Meanwhile, we observe that Dr Richardson, at the late Sanitary Congress at Leamington, entirely dissented from the theory of germs being the origin of disease, and characterised it as the wildest and most distant from the phenomena to be explained, ever conceived. As no one contests the fact that pure air is a very important factor in promoting health and averting the insidious approaches of disease, people keeping that in mind need not practically give themselves much concern about germs. See that you draw pure air into the lungs. That is an advice to which no theorist can take exception.

OCEAN-VOYAGES IN SMALL BOATS.

It is perhaps not generally known that adventurous persons occasionally cross the Atlantic from the American coast to England in small boats. The undertaking is dangerous, but is accomplished. Twenty-four years ago, when on board a Cunard steamer, our vessel passed an open sailing-boat containing two men on a voyage from America to Europe. They had no means for taking an observation, but trusted to fall in with large ships, from which they would get information as to where they were. On sighting them, our captain knew what they wanted, and hung out a black board on which were inscribed in chalk the latitude and longitude. This was satisfactory, and on they went on their perilous expedition. What came of them we know not. We were told that men who run risks of this kind, and who happen not to procure information as to their whereabouts, are apt to make strange mistakes in their voyage to England; such, for instance, as running on the coast of Spain instead of the British Islands—the whole thing a curious instance of reckless daring.

Small vessels, possibly better provided, have made runs which have attracted the admiring attention of nautical men, for the exceptional circumstances under which they occurred, but without reference to competition or bonus. In 1859 three Cornish fishermen, in a fishing-boat of small tonnage, sailed from Newlyn near Penzance to the Cape of Good Hope, and thence across the Indian Ocean to Melbourne, where they arrived 'all well.' We do not find the actual tonnage named. In 1866 a small yacht of twenty-five tons, hailing from Dublin, set out from Liverpool, and safely reached New South Wales after a run of a hundred and thirty days. The distance was set down at sixteen thousand miles. It was regarded, and justly regarded, as a bold adventure in 1874, when a schooner of only fifty-four tons safely brought over a cargo of deals from St Johns, New Brunswick, to Dublin, with but seven hands to manage the craft.

Boat-voyages, however, are evidently more remarkable than those of clippers, yachts, and schooners; on account of the extremely small dimensions of the craft which have ventured to brave the perils of the

ocean, and of the paucity of hands to manage the sails and helm during a period measured by months—under privations of various kinds.

Eleven years ago the Americans gave an indication of spirit and pluck in the conception and fulfilment of a very bold enterprise. Mr Hudson, the owner of a small craft named the *Red White and Blue*, fitted it up for an ocean-trip to England. It was a life-boat, built of galvanised iron, only twenty-six feet in length, six feet in breadth of beam, and three feet deep from deck to hold. Small as it was, the *Red White and Blue* carried what sailors call a very cloud of canvas; it had mainsail, spritsails, staysails, courses, topsails, royals, top-gallants, sternsails, trysails, three masts, bowsprit, booms, yards, gaffs, jib-boom, yard-tops, cross jack yards, spankers, and all the rest of it—an enormous amount of furniture, one would think, for so small a house. The boat was sharp at both ends, had water-tight compartments lengthwise and transverse, and safety-valves which would enable her to right herself in a few minutes if flooded. There was a tiny cockpit for the steersman near the mizzen-mast, in which he sat somewhat in the same position as Mr Macgregor in his *Rob Roy* canoe. The air-cylinders at each end of the boat and along the sides, customary in life-boats, assisted in maintaining the buoyancy and upright position. It is amusing to read of a mainmast only seven feet high and a bowsprit of two feet in length; but the juvenile ship was proportionate in all these matters, and bravely she looked, a plucky handsome little craft.

The crew of the *Red White and Blue* was as exceptional as the boat itself. The owner, Captain John M. Hudson, took the command; Mr Frank E. Fitch acted as mate; while in lieu of petty officers, able seamen, and ordinary seamen was a dog named 'Fanny.' On the 9th of July 1866 the pigmy ship took farewell of Sandy Hook, near New York, on a voyage of unknown duration and uncertain vicissitude. At midnight on the 18th the boat struck against something hard and solid, but fortunately without receiving much damage. They sailed on till the 5th of August, when they fell in with the brig *Princess Royal*, hailing from Yarmouth, and obtained a bottle of rum, two newspapers (very precious to the wayfarers), and a signal-lamp. Narrowly escaping a complete overturn on the 8th, they spoke with the barque *Welle Merryman*, from which they obtained two bottles of brandy. After another peril of capsizing, they at length sighted English land, the Bill of Portland, on the 14th. Beating up the Channel, the boat entered Margate

Harbour on the 16th, after being thirty-seven days at sea. The little craft created no small astonishment at Margate. As there was no chronometer on board, the calculations of distance, direction, &c. had to be made by compass, line, and dead-reckoning. So little opportunity had there been of obtaining a fire, that the food (mostly preserved in air-tight tins) had to be eaten cold. The original store of a hundred and twenty gallons of water supplied their wants with this essential requisite. Poor Fanny the dog did not at all relish the voyage; constant exposure to the weather so weakened her that she died soon after reaching Margate. When the *Red White and Blue* was afterwards exhibited at the Crystal Palace, a little incredulity was expressed as to the reality of the voyage; but as the names of the vessels spoken with were given and the dates of meeting, there seems no reason to doubt the faithfulness of the narrative. The two navigators, however, did not return to America in the same way; they had 'had enough of it.'

A still bolder achievement, in so far as the number of the crew was concerned, was that of Alfred Johnson, who in June 1876 started from America in a small boat manned only *by himself*. Quitting the port of Gloucester, Massachusetts, on the 15th, he had fine weather for a time, but then experienced some of the peril of Atlantic voyaging under exceptional circumstances. Fogs and head winds compelled him to put into Shake Harbour, where he had his compass corrected. Starting again on the 25th, he experienced tolerably fair weather until the 7th of July, when a heavy gale set in from the south-west. The combings of the hatchway were started, and the water, finding entrance, damaged some of his provisions. The gale subsiding, he was favoured with fine weather and fair wind until the 16th; and a strong breeze in the right direction coming on, he made good progress till the 2d of August. When about three hundred miles from the Irish coast, the wind increased to a hurricane; he hove to, but in unshipping his mast for this purpose, the boat got broadside on a large wave and was upset. Johnson clambered on the upturned bottom, where he remained for about twenty minutes. By dexterous management he succeeded in righting the boat, got in, and pumped it dry; everything, however, was wetted by the upset, and he lost his square-sail and kerosene lamp.

Wending his way as winds permitted, he reached within a hundred miles of the Irish coast by the 7th, spoke a ship, and obtained some bread and fresh water—both of which had become very scanty with him. On the following

day he got soundings, but fog prevented him from seeing land. On the 10th he sighted Milford, near the south-west extremity of Wales. He landed at Abercastle in Pembrokeshire on the 11th, after being fifty-seven days at sea; starting again, he put into Holyhead, and finally arrived at Liverpool on the 21st. The little *Centennial*, which measured only twenty feet in length over all, had run about seventy miles a day on an average. Johnson maintained his general health excellently well, though suffering from want of sleep.

The little boat that has recently crossed the Atlantic differed from Johnson's in this among other particulars, that it had a crew of two persons, one of whom was a woman. Certainly this woman will have something to talk about for the rest of her life: seeing that we may safely assign to her a position such as her sex has never before occupied—that of having managed half the navigation of a little ocean-craft for some three thousand miles. The *New Bedford*, so designated after the town of the same name in Massachusetts (the state from which Johnson also hailed), is only twenty feet long, with a burden of a little over a ton and a half; built of cedar, and rigged as (in sailor-phrases) a 'leg-of-mutton schooner;' with two masts and one anchor. Anything less ocean-like we can hardly conceive. Captain Thomas Crapo, the owner of this little affair, is an active man in the prime of life; and his better-half proves herself worthy to be the helpmate of such a man. On the 28th of May in the present year, Captain and Mrs Crapo embarked in their tiny ocean-boat, provided with such provisions and stores as they could stow away under the deck. The steersman (or steerswoman) sat in a sunken recess near the stern, with head and bust above the level of the deck; the other took any standing-place that he could get for managing the sails, rope, anchor, &c. The boat had no chronometer; and the progress had to be measured as best it could by dead-reckoning.

The boat, soon after leaving New Bedford, was forced by stress of weather to seek a few days' shelter at Chatham, a small port in the same state. Hoisting sail again on the 2d of June, the boat set off with a fair wind; and all went well for three days. An adverse wind then sprang up, a fog overspread the sky, and for ten days the voyage continued under these unfavourable circumstances. Whilst near the shoal known as the Great Banks, a keg was seen floating; this was secured, and the iron hoops utilised (with the aid of canvas) in making a drogue—one which was included among the outfit of the boat being found too light for its purpose. The boat,

after lying to for three or four days in a gale of wind, started again, and sailed on till the 21st of June, when another gale necessitated another stoppage. The *New Bedford* sighted the steamer *Batavia*, which offered to take the lonely pair of navigators on board: an offer kindly appreciated, but courteously declined. After this meeting, a succession of gales was encountered, and the rudder broke; a spare oar was made to act as a substitute. The sea ran so very high that even when lying down to rest, husband and wife had to lie on wet clothes, everything on board being sloppy and half saturated. At one portion of this trying period Captain Crapo had to steer for seventy hours uninterruptedly, his wife being incapacitated from rendering the aid which was her wont; and on another occasion he had to pay eighteen hours' close attention to the drogue. The voyage terminated on the 21st of July, after a duration of fifty-four days. The average sleep of the captain did not exceed four hours a day; and he had no sleep at all during the last seventy hours of the run. He had intended to make Falmouth his port of arrival, but was glad to make for Penzance instead.

The surname of Crapo, we were informed by the captain, is not uncommon at New Bedford. The good wife is Swedish by descent, Scotch by birth, American by marriage—a citizen of the world. In examining the boat closely (which we have done), it becomes more than ever a marvel how it could have formed the home of a married couple for seven weeks. Descending through a small hatchway, the feet rest on the floor of (let us say) the state-cabin, an apartment three feet high; consequently the head and body project above the hatchway. By spreading blankets and rugs, and crouching down by degrees, a would-be sleeper can lie down under the deck, or two sleepers close to the two sides of the boat. The wife of course acted as stewardess, cook, parlour-maid, scullery-maid, &c., leaving her husband to manage most of the navigation. The sperm-oil lamp for the compass-binnacle; the kerosene or petroleum lamp for the cooking-stove; the receptacles for biscuit and preserved meats and vegetables; the butler's pantry for a few bottles of spirits; the vessels for containing water—all were packed into a marvellously small space. The drogue (already mentioned) is a kind of floating anchor which, dragged after the vessel by means of a long rope, helps to steady it in certain states of the wind. Five hundred pounds weight of stores and six hundred of iron ballast, kept the boat sufficiently low in the water.

Such were the interior arrangements of one of those strange small vessels which adventurously attempt to cross the Atlantic.

Printed and Published by W. & R. CHAMBERS, 47 Paternoster Row, LONDON,
and 339 High Street, EDINBURGH.

All Rights Reserved.

*** END OF THE PROJECT GUTENBERG EBOOK CHAMBERS'S
JOURNAL OF POPULAR LITERATURE, SCIENCE, AND ART, NO. 724

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.